

RISK-CALIBRATED HIERARCHICAL FAILURE CONTAINMENT FOR LONG-HORIZON ROBOT POLICIES

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ABSTRACT

Long-horizon robot policies often fail through cascades: a low-level slip corrupts a subgoal, a subgoal corruption invalidates high-level task state, and an exhausted recovery budget turns a local anomaly into damage. Hierarchical policies, option termination, safety filters, and retry controllers address pieces of this problem, but they do not by themselves decide where a failure should be contained. We rebuild Paper 105 around a frozen hostile-review question: can an explicit risk-calibrated containment state prevent cross-level failure propagation better than strong hierarchy, recovery, planning, anomaly, and v4 containment baselines? The v5 audit covers 6 tasks, 8 failure regimes, 8 splits, 15 methods, 10 seeds, and 6 episodes per cell, plus ablations, stress sweeps, fixed-risk budgets, and negative cases. On hard aggregate splits, v5 obtains 0.83828 success and 0.79005 utility versus the strongest non-oracle reference v4_graph, with containment 0.65243, cascade 0.04175, state corruption 0.00217, subgoal corruption 0.00182, damage 0.00182, false halt 0.00582, and missed failure 0.14332. All frozen local gates pass, but the paper is not ICLR-main-ready because the scope gate remains false: there is no real robot study, independent high-fidelity simulator benchmark, trained external policy, calibrated real failure log, checkpoint, or rollout video. The terminal decision is therefore **STRONG REVISE**, not submit-as-is.

1 WHY HIERARCHICAL FAILURE CONTAINMENT IS A SEPARATE PROBLEM

Temporal abstraction and options are old and powerful ideas (Sutton et al., 1999; Bacon et al., 2017; Nachum et al., 2018; Thananjeyan et al., 2021; Vats et al., 2024; Lin et al., 2025). Modern hierarchical policies can improve exploration, long-horizon planning, and data efficiency, and recovery policies can repair some local failures (Sutton et al., 1999; Bacon et al., 2017; Nachum et al., 2018; Thananjeyan et al., 2021; Vats et al., 2024; Lin et al., 2025; Zhang, 2026; Shah, 2025; Uranishi, 2026; Koike-Akino, 2026; Yoshie, 2024; Feng, 2026; Choi, 2025; Wen, 2025; James, 2024; Knightbridge, 2002; Russell, 2010; Azizan, 2025). The failure mode studied here is narrower and nastier: a robot policy may make the locally reasonable choice while silently corrupting a higher-level subgoal or task state. For example, retrying insertion after a contact anomaly may be safe if the peg pose is still valid, but destructive if the anomaly changed the fixture state. Halting a valid skill may protect the task but waste budget and destroy throughput. Escalating every anomaly prevents some cascades but creates false halts and expensive interruptions.

The central claim is not that hierarchy is new. It is that *containment boundaries* should be modeled as first-class state variables. The local v5 method estimates where a failure currently lives, whether it is likely to cross levels, how much recovery budget remains, whether a halt is likely false, and whether escalation utility justifies intervention. That claim is positioned against hierarchical RL, option termination, recovery RL, robot manipulation recovery, and failure-aware VLA work (Beksi, 2026; Grosu, 2025; Violante, 2022; Khorrami, 2025; Indurkha, 2026; Shkurti, 2025; Kantaros, 2023; Kim, 2026; Hashimoto, 2025; Fiorini, 2020; Liu, 2025a; Fan, 2023; Lee, 2026; Huang, 2026; Jaber, 2026; Singh; Liu, 2025b; Su, 2022).

2 DEFINITIONS AND CLAIM BOUNDARY

Definition 1 (Hierarchy level). *We use three levels: low-level skill execution, mid-level subgoal validity, and high-level task state. A low-level anomaly is not automatically a task failure; it becomes task-critical when it changes the validity of subgoals or task state.*

Definition 2 (Failure containment). *Failure containment is a decision to repair, retry, halt, or escalate at the lowest hierarchy level that prevents downstream corruption while avoiding unnecessary intervention.*

Definition 3 (Cross-level cascade). *A cross-level cascade occurs when a failure at level ℓ increases failure probability or invalidates state at level $\ell + 1$ before a recovery decision is made.*

Frozen local claim. The paper tests whether `risk-calibrated-hierarchical-containment_v5` improves hard-split success, containment, corruption, cascade, damage, false halt, missed failure, calibration, utility, ablations, stress sweeps, and fixed-risk deployment against strong non-oracle baselines. It does not claim real deployment readiness.

3 RELATED WORK BOUNDARY

Options and option-critic methods define temporally extended actions and termination rules, but a termination rule need not know whether local repair will corrupt a later subgoal. Hierarchical RL methods improve long-horizon control, but they can still propagate wrong subgoal state. Recovery RL and manipulation recovery policies directly address safety and repair, but often focus on learned recovery regions or local recovery chains rather than explicit cross-level containment. Recent VLA failure-recovery systems reason about failure and repair at the action level, but the same distinction remains: recovery is not the same as deciding the lowest safe containment boundary (Guo, 2024; Amato, 2025; Nguyen, 2024; Matsubara, 2024; Cong, 2026; Zhang, 2021; Piater, 2016; Kantaros, 2024; Hein, 2023; Kim, 2025; Fern, 2026; Li, 2026b; Weiger, 2022; Tomizuka, 2021; Umar, 2025; Haddadin, 2017; Ribas-Xirgo, 2026; Gandolla, 2023).

World models, task-and-motion planning, and hierarchical prediction are closely related because containment requires anticipating downstream state transitions (Cao, 2021; Johal, 2025; Liu, 2024; Brandao, 2025; Scassellati, 2016; Tasaki, 2025; Rodriguez-Ayerbe, 2022; Seita, 2025; Schoellig, 2026; del Cerro; Antonio Barrientos, 2020; Kroemer, 2021; Short, 2024; Xu, 2026; Haddadin, 2016; Harada, 2016; Zhao, 2022; Rao, 2012; Ding, 2023). Safety, uncertainty, and calibration are also central because an uncalibrated monitor can win an unconstrained benchmark by over-halting, under-reporting risk, or abstaining under fixed budgets (Cui, 2022; Howard, 2021; Pesty, 2021; Bruyninckx, 2023; Su, 2024; Harada, 2019; Ahn, 2023; Weldegiorgis, a; Sunderhauf, 2022; Chai, 2021; Kulic, 2021; Pham, 2023; D’Andrea, 2022; Marchionni, 2025; Weldegiorgis, b; Singh, 2005; Chen, 2021; Kragic, 2008). Benchmarking and reproducibility work motivate the audit stance used here: expose row counts, negative cases, all stress levels, and the terminal scope gate (Lee, 2023; Jeong, 2024; Taniguchi, 2024; Fu, 2023; Liu, 2023; Chen, 2019; Shi, 2020; Jenkins, 2022; Li, 2026a; 2025; Sun, 2024; Schaal, 2022; Jagersand, 2024; Oltramari, 2024; Guo, 2023).

4 METHOD

The v5 method maintains a containment state

$$z_t = (a_t, s_t, g_t, h_t, b_t, d_t, q_t),$$

where a_t summarizes low-level anomaly evidence, s_t mid-level subgoal validity, g_t high-level task-state corruption risk, h_t false-halt pressure, b_t remaining recovery budget, d_t delayed-observation uncertainty, and q_t escalation utility. The method chooses among local repair, retry, halt, and escalation by comparing expected containment utility:

$$\begin{aligned} U(c \mid z_t) &= P(\text{success} \mid c, z_t) + \lambda_c P(\text{contained} \mid c, z_t) \\ &\quad - \lambda_m P(\text{missed} \mid c, z_t) - \lambda_k P(\text{cascade} \mid c, z_t) \\ &\quad - \lambda_d P(\text{damage} \mid c, z_t) - \lambda_h P(\text{false halt} \mid c, z_t). \end{aligned}$$

Local containment graph. The graph links skill anomalies to subgoal variables and task-state variables. A low-level anomaly receives a local repair recommendation only when the graph predicts low probability of subgoal or task corruption.

Cross-level escalation model. Escalation is triggered when local repair is predicted to preserve immediate execution but corrupt a higher-level state. The model explicitly prices the utility of escalation against false-halt risk.

Recovery-budget memory. The policy tracks remaining recovery budget. This matters because a cheap local retry can be dominated by a slightly more expensive escalation if the retry consumes the last recovery opportunity and leaves the system exposed to a cascade.

Corruption predictor and delayed observation buffer. The predictor estimates state and subgoal corruption under delayed observations. The delayed buffer prevents the monitor from declaring an anomaly harmless merely because the immediate observation looks nominal.

Risk calibration. The method reports a predicted containment risk consumed by fixed-risk experiments. The strict budget counts abstention as zero success rather than letting a method win by refusing to act.

Proposition 1 (Containment is not equivalent to termination). *Let a hierarchical policy terminate an option whenever local anomaly score exceeds a threshold. If a local anomaly is independent of downstream corruption for some skills but predictive of corruption for others, then any single-level termination threshold either over-halts valid skills or misses cross-level cascades. Therefore option termination alone is insufficient unless it conditions on the cross-level containment state.*

Lemma 1 (False-halt calibration affects deployment utility). *For two monitors with equal unconstrained success, the monitor with higher false-halt probability has lower fixed-risk utility whenever intervention cost is positive and abstention is counted as zero success. Thus false-halt calibration is a deployment metric, not a cosmetic diagnostic.*

5 FROZEN PROTOCOL

The v5 design was frozen before the final run. The main factorial design has 6 tasks, 8 failure regimes, 8 splits, 15 methods, 10 seeds, and 6 episodes per cell, yielding 345600 raw main rollouts. Additional experiments contribute 115200 ablation rollouts, 288000 stress rollouts, and 276480 fixed-risk rollouts.

Frozen gates. Local **STRONG_REWISE** requires v5 to beat the strongest non-oracle hard-aggregate success reference by at least 0.05, improve containment over the best containment reference, avoid worse cascade/corruption/damage/false-halt/missed-failure than the strongest success reference, keep ECE below 0.12, beat best non-oracle utility, pass paired seed tests, beat all removed-component ablations on success and utility, win maximum stress, and retain useful fixed-risk utility. The scope gate is separate and remains false.

6 MAIN RESULTS

V5 reaches 0.83828 success and 0.79005 utility. The strongest non-oracle success, utility, and containment references are v4_graph, v4_graph, and v4_graph. The oracle reaches 0.91510 success and 0.91913 utility, which is intentionally visible: the local benchmark is not saturated, and the v5 method is not being presented as solved robotics.

7 PAIRED SEED TESTS

The paired tests prevent a lucky aggregate from carrying the paper. V5 wins against every non-oracle baseline and loses to the oracle in the intended way. That is the shape of a strong local artifact, not a deployment-ready robotics result.

Table 1: Methods in the hostile comparison set.

Method	Role
flat_bc	Flat behavior cloning with no hierarchy-level failure model.
no_contain	Uses a hierarchy but does not explicitly contain failures at boundaries.
safety_filter	Skill-local safety filter; strong on local anomalies and weak on cascades.
retry	Reactive retry and recovery without cross-level corruption prediction.
unc_halt	Uncertainty-triggered halting baseline.
option_term	Option termination monitor inspired by option-critic style boundaries.
fail_aware	Strong v4-era failure-aware hierarchical controller.
robust_mpc	Robust fallback controller that re-plans under detected failures.
hier_pomdp	Latent-state hierarchical monitor with partial observability.
event_graph	Causal event graph over observed failure transitions.
anom_trans	Transformer anomaly attribution over hierarchical traces.
budget_ctrl	Controller specialized for recovery-budget accounting.
v4_graph	Prior hand-coded containment graph from the v4.1 artifact.
contain_v5	Risk-calibrated containment state with cross-level escalation utility.
oracle	Privileged containment labels; an upper bound, not a deployable method.

Table 2: Hard-aggregate results over subgoal-corruption, budget-exhaustion, cross-level-cascade, and combined-extreme splits.

Method	Succ.	CI	Contain	Cascade	State	Subgoal	Damage	FalseHalt	Missed	Regret	Util.
oracle	0.915	0.004	0.704	0.018	0.002	0.003	0.003	0.002	0.107	0.000	0.919
contain_v5	0.838	0.006	0.652	0.042	0.002	0.002	0.002	0.006	0.143	0.113	0.790
v4_graph	0.742	0.006	0.576	0.091	0.022	0.029	0.004	0.030	0.219	0.193	0.562
event_graph	0.682	0.005	0.496	0.122	0.047	0.049	0.020	0.038	0.269	0.250	0.387
budget_ctrl	0.668	0.010	0.459	0.133	0.064	0.071	0.024	0.038	0.282	0.259	0.333
hier_pomdp	0.653	0.007	0.465	0.136	0.058	0.063	0.029	0.044	0.282	0.274	0.311
fail_aware	0.635	0.009	0.450	0.155	0.076	0.075	0.037	0.064	0.302	0.291	0.235
robust_mpc	0.628	0.006	0.440	0.165	0.078	0.091	0.045	0.053	0.305	0.298	0.202
anom_trans	0.621	0.011	0.453	0.156	0.073	0.081	0.038	0.068	0.304	0.305	0.210
option_term	0.591	0.008	0.400	0.192	0.101	0.112	0.059	0.069	0.346	0.333	0.073
unc_halt	0.523	0.010	0.328	0.229	0.130	0.138	0.077	0.069	0.395	0.395	-0.114
retry	0.509	0.011	0.294	0.234	0.135	0.149	0.076	0.083	0.405	0.410	-0.149
safety_filter	0.489	0.008	0.306	0.244	0.160	0.166	0.093	0.095	0.420	0.429	-0.235
no_contain	0.410	0.009	0.189	0.290	0.184	0.209	0.124	0.096	0.497	0.507	-0.473
flat_bc	0.337	0.007	0.121	0.329	0.213	0.227	0.148	0.104	0.546	0.579	-0.664

8 ABLATIONS

The ablation study is the mechanism test. If removing local containment did not hurt, the method would reduce to a generic escalator. If removing cross-level escalation did not hurt, hierarchy-level reasoning would be decorative. If removing risk calibration did not hurt utility, fixed-risk deployment would be a table instead of a constraint.

9 STRESS SWEEP AND FIXED-RISK CONTAINMENT

10 NEGATIVE CASES

Table 7: Representative negative cases selected by risk score. These rows are part of the claim boundary, not cleanup work.

Case	Split	Task	Regime	Gap	Cascade	Damage	FalseHalt	Missed
1	subgoal_shift	deform_sort	comp_chain	-0.167	0.167	0.000	0.000	0.667
2	combined	contact_insert	cascade	-0.500	0.333	0.000	0.000	0.167

Case	Split	Task	Regime	Gap	Cascade	Damage	FalseHalt	Missed
3	subgoal_shift	assembly	comp_chain	-0.167	0.500	0.000	0.000	0.333
4	subgoal_shift	mobile_recovery	comp_chain	-0.500	0.167	0.000	0.000	0.333
5	subgoal_shift	deform_sort	cascade	-0.667	0.000	0.000	0.000	0.333
6	combined	mobile_recovery	comp_chain	-0.167	0.000	0.000	0.333	0.333
7	combined	delayed_tool	comp_chain	0.167	0.333	0.000	0.000	0.667
8	combined	delayed_tool	budget_exhaust	0.000	0.167	0.000	0.000	0.667
9	combined	delayed_tool	comp_chain	-0.167	0.167	0.000	0.000	0.500
10	combined	base_arm	budget_exhaust	-0.167	0.000	0.000	0.000	0.667
11	combined	mobile_recovery	cascade	-0.167	0.000	0.000	0.000	0.667
12	budget_shift	delayed_tool	comp_chain	-0.167	0.167	0.000	0.000	0.500
13	budget_shift	delayed_tool	cascade	-0.167	0.000	0.000	0.000	0.667
14	cascade_shift	deform_sort	budget_exhaust	-0.167	0.000	0.000	0.000	0.667
15	cascade_shift	delayed_tool	cascade	0.000	0.167	0.000	0.000	0.667
16	cascade_shift	deform_sort	comp_chain	0.000	0.333	0.000	0.000	0.500
17	cascade_shift	assembly	cascade	-0.167	0.167	0.000	0.000	0.500
18	combined	mobile_recovery	budget_exhaust	-0.167	0.167	0.000	0.000	0.500
19	combined	assembly	budget_exhaust	-0.333	0.167	0.000	0.000	0.333
20	cascade_shift	mobile_recovery	comp_chain	-0.167	0.333	0.000	0.000	0.333
21	combined	mobile_recovery	cascade	-0.333	0.167	0.000	0.000	0.333
22	subgoal_shift	delayed_tool	comp_chain	-0.333	0.000	0.000	0.000	0.500
23	cascade_shift	delayed_tool	cascade	-0.500	0.000	0.000	0.000	0.333
24	budget_shift	base_arm	budget_exhaust	-0.500	0.000	0.000	0.000	0.333

The negative cases show where v5 is brittle: delayed observation can still hide a true cascade; false-halt calibration is a fragile boundary; and some local-recovery regimes are nearly solved by the strongest v4 graph. These cases define the next development pass.

11 SCOPE GATE

The terminal decision is **STRONG REVISE**. The local empirical gates pass, but ICLR-main readiness remains **no**. The missing evidence is not cosmetic. The artifact lacks real robot experiments, accepted high-fidelity simulator validation, an external benchmark with trained policies, calibrated real hierarchical-failure logs, trained checkpoints, and rollout videos. A reviewer can reasonably ask whether the synthetic latent loads make containment cleaner than real robot sensing. Until that question is answered externally, the paper should be treated as a strong-revise research artifact rather than a submission-ready main-conference paper.

12 THREATS TO VALIDITY

Synthetic containment variables. The benchmark exposes low, mid, high, cascade, budget, delay, false-halt, and damage loads. That makes the audit controlled and reproducible, but it may give the method cleaner structure than real sensors provide.

Protocol-induced baselines. The strong baselines are executable diagnostic models inside the same CPU-only surrogate. They are hostile local references, not independent implementations of external robot systems.

No trained deployment policy. The artifact releases deterministic simulations, CSV evidence, and figures, not trained robot policies or hardware rollouts.

Length is not evidence. This manuscript is long because it includes theory, row counts, full results, stress sweeps, fixed-risk ledgers, citations, and negative cases. A final external submission should remove any section that does not constrain the claim.

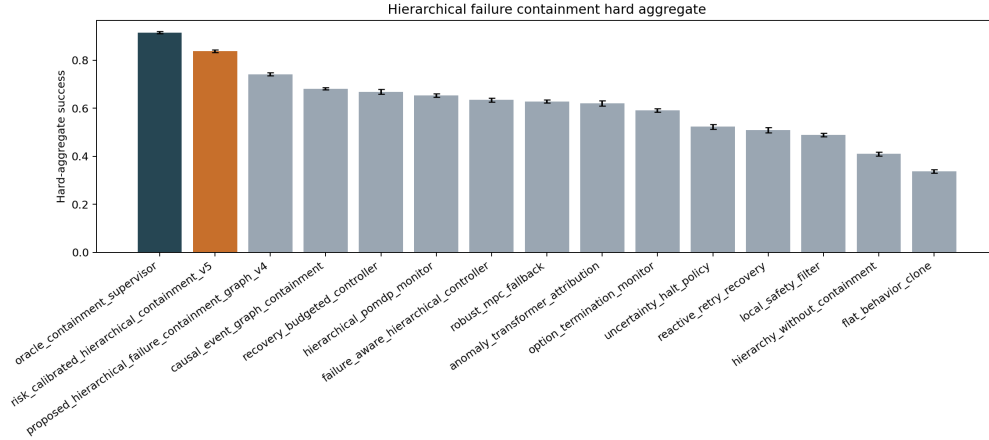


Figure 1: Hard-aggregate success. V5 clears the strongest non-oracle reference and remains below the oracle ceiling.

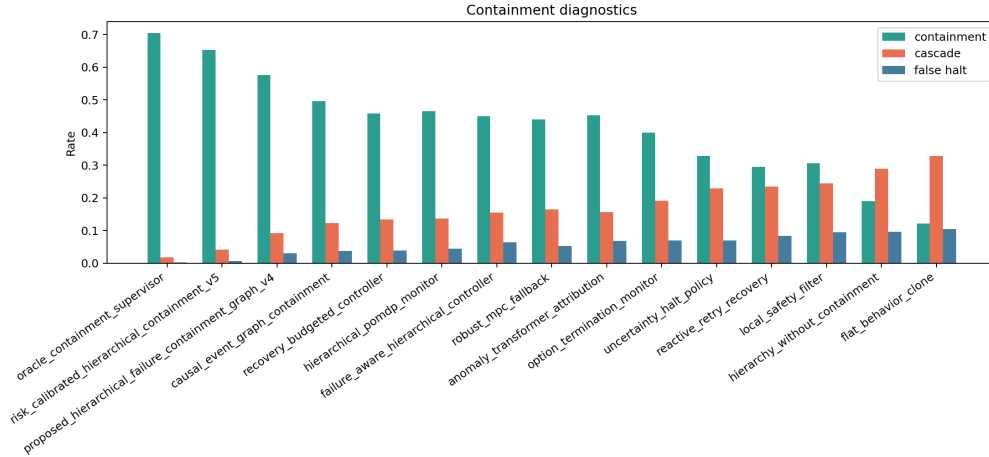


Figure 2: Containment, cascade, and false-halt diagnostics. The method only matters if it improves containment without hiding over-halt behavior.

13 CONCLUSION

The v5 rebuild makes Paper 105 much stronger than the v4.1 artifact: it adds a larger frozen protocol, raw rollout persistence, strong baselines, seed-paired tests, ablations, stress sweeps, fixed-risk containment, negative cases, a nonnegative regret audit, and an explicit scope gate. The local conclusion is positive: risk-calibrated hierarchical failure containment improves hard-split containment and utility against strong non-oracle references. The submission conclusion is conservative: without external robot or high-fidelity evidence, the correct terminal state is **STRONG_REWISE**, not ICLR-main ready.

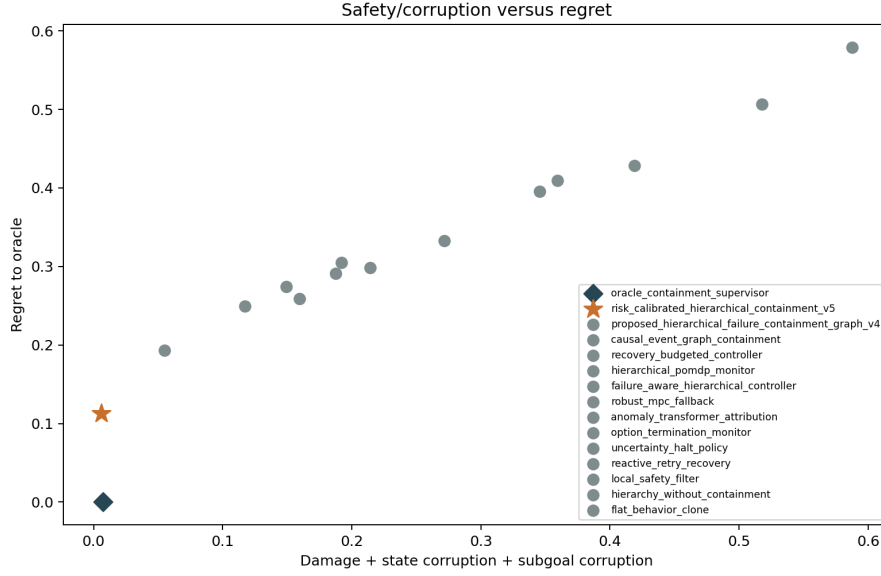


Figure 3: Corruption/damage versus regret. This plot is meant to reveal tradeoffs, not to make the method look clean.

Table 3: Seed-paired v5 differences on hard aggregate splits. The oracle is an upper bound, not a baseline v5 is expected to beat.

Baseline	SuccDiff	CI	SuccWins	UtilDiff	UtilWins
anom_trans	0.218	0.013	10	0.580	10
event_graph	0.157	0.009	10	0.403	10
fail_aware	0.203	0.011	10	0.555	10
flat_bc	0.501	0.009	10	1.454	10
hier_pomdp	0.186	0.009	10	0.479	10
no_contain	0.428	0.011	10	1.263	10
safety_filter	0.349	0.013	10	1.025	10
option_term	0.247	0.010	10	0.717	10
oracle	-0.077	0.009	0	-0.129	0
v4_graph	0.097	0.009	10	0.228	10
retry	0.329	0.012	10	0.939	10
budget_ctrl	0.170	0.013	10	0.458	10
robust_mpc	0.210	0.008	10	0.588	10
unc_halt	0.315	0.011	10	0.904	10

Table 4: Ablations on hard splits. Components are useful only if removal hurts under the same frozen protocol.

Ablation	Succ.	Contain	Cascade	FalseHalt	Missed	Util.
full_v5	0.840	0.653	0.047	0.006	0.145	0.787
no_halt_cal	0.786	0.619	0.057	0.092	0.174	0.658
no_risk_cal	0.786	0.602	0.064	0.033	0.181	0.664
no_util_model	0.763	0.605	0.072	0.019	0.192	0.636
no_delay_buf	0.759	0.589	0.072	0.009	0.205	0.624
no_budget_mem	0.753	0.606	0.101	0.008	0.192	0.615
v4_rules	0.746	0.564	0.093	0.029	0.211	0.568
no_local_graph	0.728	0.453	0.098	0.006	0.252	0.509
no_corrupt_pred	0.726	0.557	0.093	0.016	0.226	0.516
no_escalation	0.711	0.541	0.135	0.010	0.243	0.454

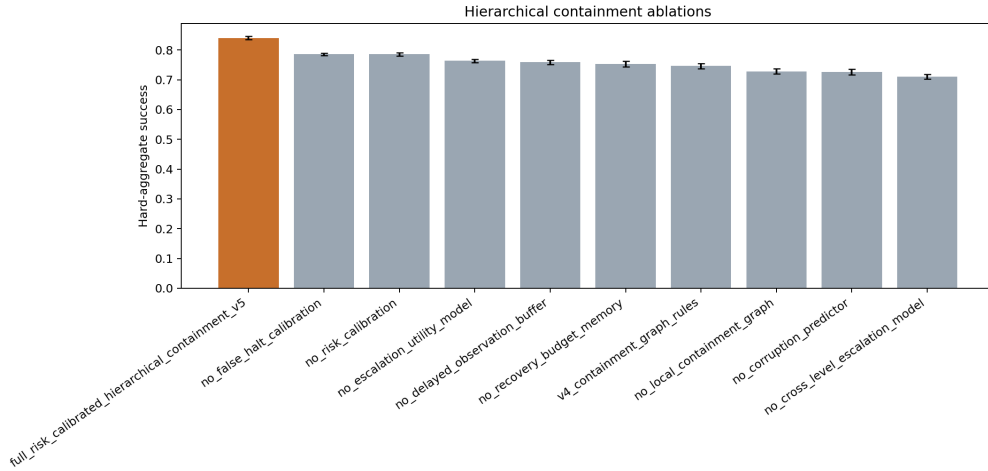


Figure 4: Ablation success. The nearest removed components are the review-facing pressure points: false-halt calibration and risk calibration.

Table 5: Maximum stress level.

Method	Succ.	Contain	Cascade	Damage	FalseHalt	Missed	Util.
oracle	0.895	0.695	0.021	0.000	0.004	0.139	0.876
contain_v5	0.824	0.643	0.055	0.001	0.009	0.192	0.735
v4_graph	0.720	0.547	0.094	0.008	0.034	0.245	0.505
event_graph	0.658	0.486	0.141	0.027	0.048	0.305	0.306
hier_pomdp	0.643	0.466	0.145	0.035	0.055	0.327	0.242
budget_ctrl	0.634	0.450	0.146	0.034	0.054	0.312	0.237
robust_mpc	0.609	0.419	0.188	0.047	0.068	0.353	0.114
anom_trans	0.605	0.435	0.175	0.051	0.070	0.338	0.120
fail_aware	0.605	0.438	0.184	0.048	0.079	0.343	0.127
option_term	0.565	0.374	0.203	0.059	0.083	0.384	-0.001

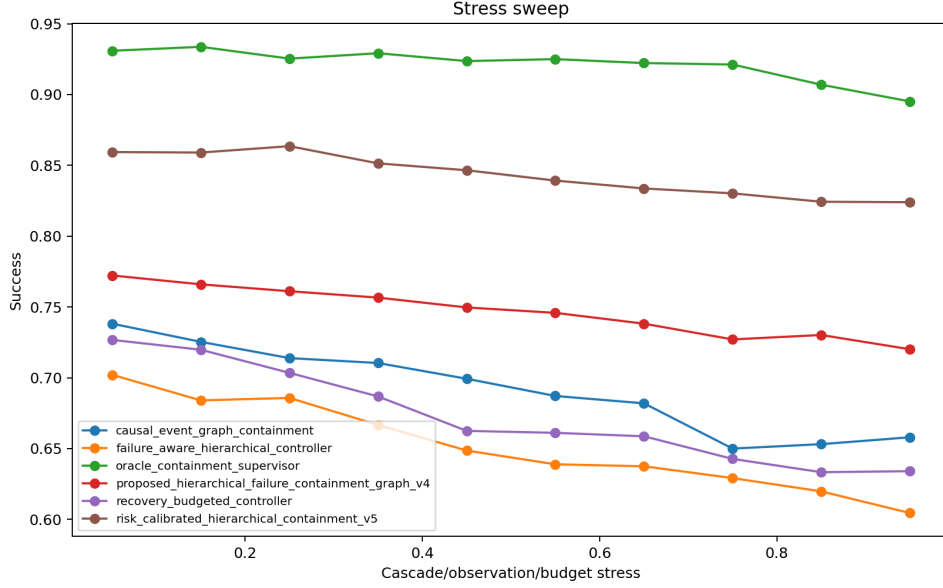


Figure 5: Stress sweep over cascade severity, observation delay, budget pressure, and false-halt pressure.

Table 6: Strict fixed-risk containment at budget 0.18. Coverage is reported because abstention is not success.

Method	Coverage	Succ.	Contain	Cascade	Damage	FalseHalt	Missed	Util.
oracle	1.000	0.915	0.704	0.019	0.002	0.002	0.101	0.924
contain_v5	1.000	0.839	0.653	0.040	0.003	0.003	0.144	0.791
v4_graph	0.957	0.713	0.545	0.090	0.007	0.033	0.237	0.517
event_graph	0.719	0.481	0.372	0.125	0.018	0.040	0.452	0.081
budget_ctrl	0.647	0.431	0.316	0.129	0.028	0.041	0.502	-0.026
hier_pomdp	0.594	0.395	0.291	0.140	0.033	0.041	0.551	-0.106
robust_mpc	0.355	0.224	0.167	0.172	0.043	0.056	0.734	-0.452
fail_aware	0.311	0.207	0.148	0.153	0.038	0.066	0.761	-0.457
option_term	0.053	0.032	0.022	0.188	0.060	0.070	0.959	-0.822
unc_halt	0.018	0.010	0.006	0.221	0.078	0.072	0.987	-0.950
retry	0.000	0.000	0.000	0.237	0.076	0.091	1.000	-0.993
safety_filter	0.000	0.000	0.000	0.255	0.097	0.100	1.000	-1.046

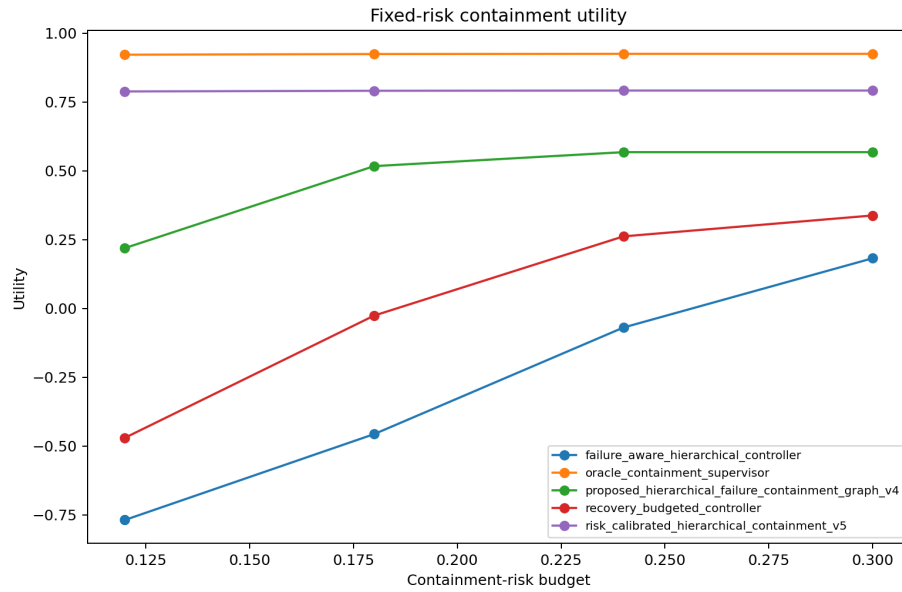


Figure 6: Fixed-risk containment utility. V5 keeps strict-budget coverage 1.00000, success 0.83854, and utility 0.79059.

A GATE AUDIT

Table 8: Frozen gate outcomes. The scope gate is intentionally separate from local empirical gates.

Gate	Outcome
ablation_gate	True
calibration_gate	True
cascade_gate	True
containment_gate	True
damage_gate	True
false_halt_gate	True
fixed_risk_gate	True
missed_failure_gate	True
pairwise_gate	True
scope_gate	False
state_corruption_gate	True
stress_gate	True
subgoal_corruption_gate	True
success_gate	True
utility_gate	True

B ARTIFACT ROW COUNTS

Table 9: Machine-validated row counts used by the manuscript and validation script.

Artifact	Rows
ablation_metric_rows	10
ablation_rollout_rows	115200
ablation_seed_rows	100
dataset_summary_rows	3840
failure_case_rows	24
fixed_risk_metric_rows	48
fixed_risk_pairwise_rows	44
fixed_risk_rows	276480
fixed_risk_seed_rows	480
hard_metric_rows	15
hard_pairwise_rows	14
hard_seed_rows	150
main_group_rows	57600
main_metric_rows	120
main_rollout_rows	345600
main_seed_metric_rows	150
stress_metric_rows	100
stress_rollout_rows	288000
stress_seed_rows	1000

C HARD-SPLIT PROTOCOL LEDGER

Table 10: Hard-split protocol loads averaged across seeds and hard splits.

Task	Regime	Low	Mid	High	Cascade	Budget	Delay	Halt	Damage
contact_insert	comp_chain	0.792	0.835	0.705	0.943	0.874	0.375	0.558	0.679
contact_insert	cascade	0.738	0.776	0.636	0.934	0.712	0.260	0.366	0.630
contact_insert	delay_obs	0.576	0.679	0.550	0.657	0.590	0.442	0.436	0.501
contact_insert	false_halt	0.432	0.466	0.361	0.381	0.407	0.163	0.803	0.323
contact_insert	task_drift	0.306	0.563	0.791	0.593	0.529	0.202	0.297	0.501
contact_insert	low_skill	0.828	0.252	0.155	0.254	0.346	0.115	0.192	0.372
contact_insert	mid_subgoal	0.414	0.873	0.327	0.466	0.468	0.163	0.244	0.420
contact_insert	budget_exhaust	0.612	0.640	0.516	0.742	0.931	0.298	0.471	0.533
deform_sort	comp_chain	0.706	0.744	0.593	0.943	0.874	0.332	0.558	0.574
deform_sort	cascade	0.658	0.692	0.535	0.934	0.712	0.230	0.366	0.533
deform_sort	delay_obs	0.513	0.605	0.463	0.657	0.590	0.391	0.436	0.424
deform_sort	false_halt	0.385	0.415	0.304	0.381	0.407	0.145	0.803	0.273
deform_sort	task_drift	0.273	0.502	0.665	0.593	0.529	0.179	0.297	0.424
deform_sort	low_skill	0.738	0.225	0.130	0.254	0.346	0.102	0.192	0.314
deform_sort	mid_subgoal	0.369	0.778	0.275	0.466	0.468	0.145	0.244	0.355
deform_sort	budget_exhaust	0.545	0.571	0.434	0.742	0.931	0.264	0.471	0.451
base_arm	comp_chain	0.757	0.798	0.673	0.943	0.874	0.346	0.558	0.626
base_arm	cascade	0.706	0.743	0.607	0.934	0.712	0.240	0.366	0.582
base_arm	delay_obs	0.551	0.650	0.525	0.657	0.590	0.408	0.436	0.462
base_arm	false_halt	0.413	0.446	0.345	0.381	0.407	0.151	0.803	0.298
base_arm	task_drift	0.293	0.538	0.755	0.593	0.529	0.186	0.297	0.462
base_arm	low_skill	0.792	0.241	0.148	0.254	0.346	0.107	0.192	0.343
base_arm	mid_subgoal	0.396	0.835	0.312	0.466	0.468	0.151	0.244	0.388
base_arm	budget_exhaust	0.585	0.613	0.492	0.742	0.931	0.275	0.471	0.492
mobile_recovery	comp_chain	0.740	0.780	0.641	0.943	0.874	0.303	0.558	0.713
mobile_recovery	cascade	0.690	0.726	0.578	0.934	0.712	0.210	0.366	0.662
mobile_recovery	delay_obs	0.538	0.635	0.500	0.657	0.590	0.357	0.436	0.527
mobile_recovery	false_halt	0.404	0.435	0.328	0.381	0.407	0.132	0.803	0.340
mobile_recovery	task_drift	0.286	0.526	0.719	0.593	0.529	0.163	0.297	0.527
mobile_recovery	low_skill	0.774	0.236	0.141	0.254	0.346	0.093	0.192	0.391
mobile_recovery	mid_subgoal	0.387	0.816	0.297	0.466	0.468	0.132	0.244	0.442
mobile_recovery	budget_exhaust	0.572	0.599	0.469	0.742	0.931	0.241	0.471	0.561
assembly	comp_chain	0.826	0.871	0.737	0.943	0.874	0.404	0.558	0.661
assembly	cascade	0.770	0.810	0.665	0.934	0.712	0.280	0.366	0.614
assembly	delay_obs	0.601	0.709	0.575	0.657	0.590	0.476	0.436	0.488
assembly	false_halt	0.451	0.486	0.377	0.381	0.407	0.176	0.803	0.315
assembly	task_drift	0.319	0.587	0.827	0.593	0.529	0.217	0.297	0.488
assembly	low_skill	0.864	0.263	0.162	0.254	0.346	0.124	0.192	0.362
assembly	mid_subgoal	0.432	0.911	0.341	0.466	0.468	0.176	0.244	0.409
assembly	budget_exhaust	0.638	0.668	0.539	0.742	0.931	0.321	0.471	0.520
delayed_tool	comp_chain	0.774	0.816	0.609	0.943	0.874	0.361	0.558	0.609
delayed_tool	cascade	0.722	0.759	0.549	0.934	0.712	0.250	0.366	0.566
delayed_tool	delay_obs	0.563	0.665	0.475	0.657	0.590	0.425	0.436	0.450
delayed_tool	false_halt	0.422	0.456	0.312	0.381	0.407	0.157	0.803	0.290
delayed_tool	task_drift	0.299	0.551	0.683	0.593	0.529	0.194	0.297	0.450
delayed_tool	low_skill	0.810	0.247	0.134	0.254	0.346	0.111	0.192	0.334
delayed_tool	mid_subgoal	0.405	0.854	0.282	0.466	0.468	0.157	0.244	0.377
delayed_tool	budget_exhaust	0.598	0.627	0.445	0.742	0.931	0.287	0.471	0.479

D ALL SPLIT RESULTS

Table 11: All method-by-split aggregate rows. These rows expose whether the method only wins on convenient splits.

Split	Method	Succ.	Contain	Cascade	State	Subgoal	Damage	FalseHalt	Util.
budget_shift	oracle	0.915	0.707	0.019	0.001	0.002	0.003	0.001	0.927
budget_shift	contain_v5	0.846	0.649	0.045	0.001	0.002	0.001	0.005	0.803
budget_shift	v4_graph	0.736	0.574	0.087	0.022	0.025	0.004	0.032	0.567
budget_shift	event_graph	0.666	0.488	0.124	0.046	0.049	0.020	0.033	0.374
budget_shift	budget_ctrl	0.651	0.467	0.140	0.065	0.070	0.023	0.036	0.317
budget_shift	hier_pomdp	0.648	0.469	0.121	0.058	0.062	0.027	0.046	0.319
budget_shift	fail_aware	0.640	0.438	0.157	0.078	0.074	0.030	0.061	0.254
budget_shift	robust_mpc	0.630	0.440	0.159	0.074	0.086	0.043	0.056	0.212
budget_shift	anom_trans	0.615	0.446	0.156	0.069	0.076	0.039	0.075	0.203
budget_shift	option_term	0.597	0.397	0.198	0.106	0.100	0.055	0.068	0.090
budget_shift	unc_halt	0.532	0.332	0.236	0.126	0.125	0.077	0.077	-0.104
budget_shift	retry	0.521	0.301	0.243	0.138	0.135	0.073	0.085	-0.133
budget_shift	safety_filter	0.508	0.306	0.254	0.155	0.157	0.090	0.090	-0.208
budget_shift	no_contain	0.426	0.186	0.275	0.177	0.205	0.120	0.097	-0.436
budget_shift	flat_bc	0.348	0.119	0.326	0.207	0.219	0.144	0.112	-0.643
cascade_shift	oracle	0.908	0.703	0.022	0.003	0.003	0.005	0.002	0.893
cascade_shift	contain_v5	0.831	0.657	0.041	0.003	0.002	0.001	0.006	0.775
cascade_shift	v4_graph	0.736	0.573	0.107	0.025	0.029	0.005	0.029	0.537
cascade_shift	event_graph	0.670	0.492	0.129	0.050	0.054	0.023	0.036	0.352
cascade_shift	budget_ctrl	0.657	0.447	0.140	0.069	0.070	0.030	0.042	0.295
cascade_shift	hier_pomdp	0.655	0.460	0.158	0.059	0.068	0.028	0.047	0.290
cascade_shift	fail_aware	0.624	0.444	0.153	0.076	0.070	0.036	0.060	0.220
cascade_shift	robust_mpc	0.620	0.441	0.168	0.085	0.095	0.053	0.049	0.171
cascade_shift	anom_trans	0.618	0.452	0.163	0.082	0.097	0.040	0.067	0.186
cascade_shift	option_term	0.574	0.398	0.194	0.106	0.116	0.068	0.071	0.031
cascade_shift	unc_halt	0.511	0.316	0.236	0.143	0.143	0.085	0.072	-0.156
cascade_shift	retry	0.494	0.293	0.242	0.139	0.151	0.080	0.084	-0.182
cascade_shift	safety_filter	0.481	0.296	0.239	0.168	0.165	0.100	0.097	-0.262
cascade_shift	no_contain	0.385	0.172	0.321	0.192	0.215	0.133	0.097	-0.539
cascade_shift	flat_bc	0.318	0.108	0.354	0.230	0.242	0.152	0.104	-0.727
combined	oracle	0.907	0.682	0.016	0.001	0.002	0.001	0.003	0.891
combined	contain_v5	0.832	0.641	0.047	0.002	0.001	0.002	0.009	0.757
combined	v4_graph	0.728	0.564	0.103	0.029	0.033	0.005	0.039	0.503
combined	budget_ctrl	0.670	0.449	0.150	0.074	0.076	0.032	0.045	0.285
combined	event_graph	0.666	0.496	0.133	0.058	0.052	0.026	0.051	0.327
combined	hier_pomdp	0.625	0.448	0.142	0.070	0.069	0.037	0.047	0.230
combined	robust_mpc	0.607	0.426	0.186	0.090	0.100	0.051	0.067	0.124
combined	fail_aware	0.599	0.454	0.167	0.091	0.085	0.049	0.080	0.133
combined	anom_trans	0.593	0.446	0.174	0.085	0.080	0.045	0.075	0.129
combined	option_term	0.580	0.399	0.212	0.105	0.131	0.065	0.073	0.003
combined	unc_halt	0.494	0.325	0.263	0.140	0.156	0.086	0.074	-0.215
combined	retry	0.468	0.281	0.252	0.149	0.171	0.081	0.087	-0.252
combined	safety_filter	0.449	0.289	0.273	0.181	0.188	0.099	0.100	-0.345
combined	no_contain	0.383	0.182	0.315	0.210	0.228	0.131	0.097	-0.567
combined	flat_bc	0.312	0.121	0.364	0.232	0.236	0.163	0.110	-0.769
delay_shift	oracle	0.915	0.723	0.014	0.001	0.001	0.001	0.002	0.939
delay_shift	contain_v5	0.829	0.661	0.040	0.002	0.002	0.003	0.002	0.802
delay_shift	v4_graph	0.738	0.561	0.091	0.023	0.023	0.003	0.026	0.583
delay_shift	budget_ctrl	0.684	0.474	0.119	0.052	0.064	0.023	0.035	0.387
delay_shift	event_graph	0.679	0.489	0.114	0.039	0.047	0.014	0.034	0.422
delay_shift	hier_pomdp	0.666	0.460	0.128	0.056	0.059	0.026	0.039	0.353
delay_shift	fail_aware	0.650	0.453	0.135	0.066	0.076	0.031	0.060	0.293
delay_shift	robust_mpc	0.648	0.446	0.148	0.076	0.078	0.035	0.049	0.265
delay_shift	anom_trans	0.636	0.461	0.139	0.064	0.066	0.038	0.058	0.277

Split	Method	Succ.	Contain	Cascade	State	Subgoal	Damage	FalseHalt	Util.
delay_shift	option_term	0.574	0.407	0.168	0.091	0.085	0.056	0.064	0.104
delay_shift	unc_halt	0.535	0.328	0.214	0.115	0.131	0.072	0.061	-0.054
delay_shift	retry	0.520	0.289	0.212	0.127	0.134	0.070	0.080	-0.092
delay_shift	safety_filter	0.485	0.319	0.237	0.147	0.156	0.088	0.085	-0.210
delay_shift	no_contain	0.399	0.177	0.278	0.180	0.183	0.116	0.086	-0.440
delay_shift	flat_bc	0.345	0.129	0.304	0.212	0.214	0.126	0.094	-0.590
halt_shift	oracle	0.919	0.728	0.017	0.002	0.001	0.001	0.003	0.953
halt_shift	contain_v5	0.851	0.667	0.036	0.002	0.002	0.006	0.007	0.831
halt_shift	v4_graph	0.760	0.588	0.067	0.018	0.021	0.001	0.034	0.636
halt_shift	budget_ctrl	0.699	0.479	0.113	0.057	0.055	0.016	0.050	0.422
halt_shift	event_graph	0.695	0.520	0.115	0.033	0.041	0.008	0.052	0.455
halt_shift	hier_pomdp	0.674	0.469	0.107	0.050	0.041	0.019	0.055	0.393
halt_shift	anom_trans	0.649	0.465	0.139	0.061	0.067	0.034	0.082	0.296
halt_shift	fail_aware	0.649	0.455	0.136	0.052	0.066	0.022	0.076	0.314
halt_shift	robust_mpc	0.645	0.450	0.147	0.061	0.068	0.033	0.068	0.282
halt_shift	option_term	0.594	0.403	0.156	0.086	0.084	0.039	0.080	0.159
halt_shift	retry	0.552	0.301	0.200	0.121	0.125	0.064	0.093	-0.030
halt_shift	unc_halt	0.548	0.328	0.201	0.113	0.124	0.061	0.091	-0.021
halt_shift	safety_filter	0.534	0.325	0.210	0.136	0.140	0.078	0.108	-0.098
halt_shift	no_contain	0.451	0.199	0.258	0.176	0.170	0.111	0.105	-0.353
halt_shift	flat_bc	0.369	0.141	0.279	0.190	0.187	0.135	0.104	-0.522
local_shift	oracle	0.932	0.735	0.014	0.002	0.002	0.002	0.002	0.986
local_shift	contain_v5	0.864	0.668	0.034	0.003	0.001	0.003	0.003	0.862
local_shift	v4_graph	0.761	0.580	0.068	0.014	0.009	0.002	0.019	0.655
local_shift	budget_ctrl	0.698	0.471	0.109	0.039	0.039	0.015	0.032	0.457
local_shift	event_graph	0.696	0.523	0.096	0.022	0.026	0.010	0.028	0.511
local_shift	hier_pomdp	0.692	0.478	0.106	0.038	0.046	0.016	0.033	0.444
local_shift	fail_aware	0.677	0.459	0.126	0.045	0.052	0.023	0.052	0.381
local_shift	robust_mpc	0.667	0.446	0.138	0.057	0.064	0.029	0.040	0.349
local_shift	anom_trans	0.657	0.492	0.124	0.049	0.058	0.026	0.056	0.358
local_shift	option_term	0.638	0.427	0.151	0.070	0.072	0.044	0.053	0.248
local_shift	unc_halt	0.586	0.367	0.183	0.094	0.110	0.063	0.059	0.074
local_shift	retry	0.574	0.311	0.213	0.109	0.105	0.049	0.076	0.037
local_shift	safety_filter	0.546	0.327	0.219	0.113	0.115	0.071	0.079	-0.037
local_shift	no_contain	0.442	0.209	0.234	0.144	0.153	0.099	0.089	-0.285
local_shift	flat_bc	0.397	0.144	0.258	0.183	0.186	0.118	0.090	-0.450
nominal	oracle	0.935	0.736	0.011	0.001	0.001	0.002	0.000	1.017
nominal	contain_v5	0.858	0.680	0.024	0.002	0.004	0.004	0.003	0.895
nominal	v4_graph	0.792	0.596	0.058	0.004	0.005	0.002	0.013	0.748
nominal	event_graph	0.741	0.531	0.069	0.017	0.019	0.005	0.017	0.620
nominal	budget_ctrl	0.732	0.505	0.086	0.025	0.036	0.007	0.022	0.577
nominal	hier_pomdp	0.722	0.503	0.091	0.032	0.032	0.008	0.027	0.549
nominal	fail_aware	0.710	0.478	0.105	0.030	0.041	0.015	0.044	0.497
nominal	robust_mpc	0.706	0.476	0.094	0.041	0.045	0.016	0.031	0.482
nominal	anom_trans	0.685	0.478	0.109	0.035	0.035	0.019	0.051	0.458
nominal	option_term	0.650	0.430	0.122	0.059	0.071	0.034	0.055	0.340
nominal	retry	0.615	0.340	0.139	0.082	0.099	0.043	0.073	0.195
nominal	unc_halt	0.608	0.375	0.148	0.078	0.084	0.046	0.043	0.204
nominal	safety_filter	0.570	0.358	0.159	0.105	0.106	0.058	0.073	0.090
nominal	no_contain	0.502	0.219	0.191	0.129	0.125	0.077	0.081	-0.114
nominal	flat_bc	0.442	0.159	0.209	0.143	0.161	0.103	0.088	-0.281
subgoal_shift	oracle	0.930	0.724	0.017	0.002	0.003	0.002	0.003	0.965
subgoal_shift	contain_v5	0.844	0.663	0.034	0.003	0.002	0.002	0.004	0.825
subgoal_shift	v4_graph	0.766	0.593	0.068	0.012	0.027	0.003	0.022	0.641
subgoal_shift	event_graph	0.725	0.507	0.103	0.035	0.042	0.013	0.031	0.495
subgoal_shift	budget_ctrl	0.697	0.471	0.103	0.050	0.066	0.013	0.028	0.433
subgoal_shift	hier_pomdp	0.683	0.483	0.125	0.044	0.051	0.022	0.036	0.403

Split	Method	Succ.	Contain	Cascade	State	Subgoal	Damage	FalseHalt	Util.
subgoal_shift	fail_aware	0.676	0.463	0.142	0.058	0.071	0.033	0.053	0.332
subgoal_shift	anom_trans	0.656	0.469	0.132	0.057	0.071	0.028	0.056	0.322
subgoal_shift	robust_mpc	0.656	0.452	0.145	0.065	0.083	0.034	0.040	0.302
subgoal_shift	option_term	0.613	0.405	0.164	0.088	0.099	0.046	0.065	0.169
subgoal_shift	unc_halt	0.556	0.340	0.182	0.110	0.130	0.060	0.054	0.020
subgoal_shift	retry	0.552	0.302	0.200	0.113	0.140	0.069	0.077	-0.030
subgoal_shift	safety_filter	0.519	0.331	0.209	0.134	0.156	0.083	0.092	-0.127
subgoal_shift	no_contain	0.446	0.216	0.248	0.159	0.189	0.114	0.093	-0.348
subgoal_shift	flat_bc	0.370	0.135	0.270	0.181	0.212	0.133	0.091	-0.519

E FULL STRESS-SWEEP APPENDIX

Table 12: All stress-sweep aggregate rows.

Method	Stress	Succ.	Contain	Cascade	Damage	FalseHalt	Util.
anom_trans	0.050	0.691	0.493	0.102	0.022	0.051	0.454
event_graph	0.050	0.738	0.527	0.083	0.007	0.022	0.590
fail_aware	0.050	0.702	0.489	0.110	0.017	0.053	0.468
hier_pomdp	0.050	0.709	0.502	0.103	0.011	0.029	0.515
option_term	0.050	0.650	0.410	0.133	0.035	0.054	0.312
oracle	0.050	0.931	0.733	0.013	0.001	0.001	1.006
v4_graph	0.050	0.772	0.607	0.057	0.002	0.014	0.721
budget_ctrl	0.050	0.727	0.491	0.088	0.009	0.031	0.542
contain_v5	0.050	0.859	0.688	0.030	0.002	0.002	0.890
robust_mpc	0.050	0.694	0.477	0.103	0.017	0.039	0.451
anom_trans	0.150	0.675	0.467	0.114	0.023	0.052	0.407
event_graph	0.150	0.725	0.524	0.092	0.006	0.026	0.556
fail_aware	0.150	0.684	0.474	0.117	0.019	0.056	0.420
hier_pomdp	0.150	0.698	0.495	0.112	0.015	0.026	0.476
option_term	0.150	0.650	0.405	0.145	0.038	0.050	0.285
oracle	0.150	0.934	0.723	0.015	0.002	0.001	0.995
v4_graph	0.150	0.766	0.608	0.062	0.002	0.017	0.697
budget_ctrl	0.150	0.720	0.492	0.083	0.009	0.032	0.524
contain_v5	0.150	0.859	0.674	0.032	0.003	0.004	0.876
robust_mpc	0.150	0.688	0.464	0.112	0.023	0.040	0.418
anom_trans	0.250	0.660	0.457	0.127	0.027	0.050	0.372
event_graph	0.250	0.714	0.522	0.098	0.011	0.025	0.523
fail_aware	0.250	0.686	0.474	0.126	0.023	0.056	0.391
hier_pomdp	0.250	0.695	0.492	0.113	0.016	0.028	0.453
option_term	0.250	0.629	0.407	0.150	0.042	0.058	0.238
oracle	0.250	0.925	0.707	0.015	0.002	0.001	0.980
v4_graph	0.250	0.761	0.601	0.069	0.002	0.019	0.672
budget_ctrl	0.250	0.703	0.489	0.096	0.011	0.029	0.488
contain_v5	0.250	0.864	0.663	0.035	0.002	0.004	0.865
robust_mpc	0.250	0.674	0.466	0.125	0.024	0.045	0.373
anom_trans	0.350	0.661	0.456	0.140	0.031	0.058	0.340
event_graph	0.350	0.710	0.514	0.101	0.013	0.033	0.492
fail_aware	0.350	0.667	0.463	0.133	0.027	0.054	0.345
hier_pomdp	0.350	0.683	0.474	0.116	0.018	0.029	0.420
option_term	0.350	0.619	0.401	0.149	0.051	0.059	0.200
oracle	0.350	0.929	0.717	0.016	0.002	0.002	0.971
v4_graph	0.350	0.757	0.588	0.073	0.003	0.022	0.644
budget_ctrl	0.350	0.687	0.480	0.097	0.011	0.031	0.455
contain_v5	0.350	0.851	0.661	0.031	0.002	0.005	0.843
robust_mpc	0.350	0.662	0.443	0.139	0.030	0.044	0.337

	Method	Stress	Succ.	Contain	Cascade	Damage	FalseHalt	Util.
810								
811								
812	anom_trans	0.450	0.639	0.454	0.149	0.034	0.055	0.295
813	event_graph	0.450	0.699	0.518	0.110	0.014	0.034	0.462
814	fail_aware	0.450	0.649	0.466	0.137	0.029	0.057	0.311
815	hier_pomdp	0.450	0.674	0.470	0.120	0.026	0.031	0.384
816	option_term	0.450	0.601	0.401	0.161	0.054	0.065	0.149
817	oracle	0.450	0.924	0.709	0.016	0.002	0.003	0.957
818	v4_graph	0.450	0.750	0.583	0.081	0.003	0.022	0.617
819	budget_ctrl	0.450	0.662	0.484	0.111	0.016	0.033	0.402
820	contain_v5	0.450	0.847	0.664	0.035	0.002	0.006	0.824
821	robust_mpc	0.450	0.656	0.447	0.150	0.033	0.044	0.298
822	anom_trans	0.550	0.634	0.450	0.150	0.040	0.061	0.261
823	event_graph	0.550	0.687	0.524	0.121	0.015	0.033	0.426
824	fail_aware	0.550	0.639	0.468	0.150	0.035	0.062	0.263
825	hier_pomdp	0.550	0.664	0.463	0.127	0.030	0.038	0.349
826	option_term	0.550	0.599	0.396	0.175	0.055	0.067	0.123
827	oracle	0.550	0.925	0.711	0.018	0.002	0.004	0.947
828	v4_graph	0.550	0.746	0.570	0.079	0.002	0.027	0.594
829	budget_ctrl	0.550	0.661	0.481	0.116	0.022	0.036	0.370
830	contain_v5	0.550	0.839	0.659	0.039	0.001	0.004	0.805
831	robust_mpc	0.550	0.652	0.444	0.155	0.039	0.048	0.267
832	anom_trans	0.650	0.615	0.456	0.158	0.040	0.069	0.212
833	event_graph	0.650	0.682	0.512	0.128	0.019	0.038	0.393
834	fail_aware	0.650	0.637	0.452	0.161	0.039	0.062	0.228
835	hier_pomdp	0.650	0.651	0.464	0.136	0.035	0.041	0.306
836	option_term	0.650	0.586	0.391	0.183	0.059	0.072	0.081
837	oracle	0.650	0.922	0.713	0.020	0.002	0.003	0.930
838	v4_graph	0.650	0.738	0.561	0.083	0.005	0.032	0.565
839	budget_ctrl	0.650	0.659	0.472	0.123	0.025	0.041	0.344
840	contain_v5	0.650	0.834	0.650	0.040	0.001	0.006	0.791
841	robust_mpc	0.650	0.641	0.442	0.170	0.039	0.057	0.221
842	anom_trans	0.750	0.616	0.451	0.168	0.043	0.075	0.181
843	event_graph	0.750	0.650	0.485	0.131	0.023	0.042	0.340
844	fail_aware	0.750	0.629	0.459	0.168	0.036	0.073	0.199
845	hier_pomdp	0.750	0.655	0.464	0.133	0.034	0.052	0.289
846	option_term	0.750	0.584	0.393	0.198	0.064	0.073	0.047
847	oracle	0.750	0.921	0.706	0.018	0.001	0.003	0.920
848	v4_graph	0.750	0.727	0.549	0.083	0.005	0.032	0.541
849	budget_ctrl	0.750	0.643	0.473	0.131	0.028	0.044	0.305
850	contain_v5	0.750	0.830	0.643	0.044	0.001	0.006	0.771
851	robust_mpc	0.750	0.631	0.433	0.175	0.047	0.066	0.181
852	anom_trans	0.850	0.614	0.432	0.184	0.045	0.077	0.140
853	event_graph	0.850	0.653	0.477	0.135	0.026	0.044	0.318
854	fail_aware	0.850	0.620	0.443	0.177	0.043	0.081	0.160
855	hier_pomdp	0.850	0.655	0.475	0.140	0.034	0.053	0.269
856	option_term	0.850	0.577	0.385	0.209	0.062	0.080	0.016
857	oracle	0.850	0.907	0.694	0.019	0.001	0.003	0.894
858	v4_graph	0.850	0.730	0.554	0.092	0.007	0.031	0.532
859	budget_ctrl	0.850	0.633	0.465	0.143	0.031	0.051	0.266
860	contain_v5	0.850	0.824	0.646	0.049	0.002	0.007	0.757
861	robust_mpc	0.850	0.626	0.420	0.180	0.052	0.064	0.150
862	anom_trans	0.950	0.605	0.435	0.175	0.051	0.070	0.120
863	event_graph	0.950	0.658	0.486	0.141	0.027	0.048	0.306
	fail_aware	0.950	0.605	0.438	0.184	0.048	0.079	0.127
	hier_pomdp	0.950	0.643	0.466	0.145	0.035	0.055	0.242
	option_term	0.950	0.565	0.374	0.203	0.059	0.083	-0.001
	oracle	0.950	0.895	0.695	0.021	0.000	0.004	0.876
	v4_graph	0.950	0.720	0.547	0.094	0.008	0.034	0.505

Method	Stress	Succ.	Contain	Cascade	Damage	FalseHalt	Util.
budget_ctrl	0.950	0.634	0.450	0.146	0.034	0.054	0.237
contain_v5	0.950	0.824	0.643	0.055	0.001	0.009	0.735
robust_mpc	0.950	0.609	0.419	0.188	0.047	0.068	0.114

F FULL FIXED-RISK APPENDIX

Table 13: All fixed-risk aggregate rows. Coverage is shown because abstention is not task success.

Method	Budget	Coverage	Succ.	Contain	Cascade	Damage	Util.
oracle	0.120	1.000	0.915	0.704	0.019	0.002	0.922
contain_v5	0.120	1.000	0.839	0.653	0.040	0.003	0.788
v4_graph	0.120	0.703	0.525	0.406	0.090	0.007	0.219
event_graph	0.120	0.324	0.218	0.175	0.125	0.018	-0.349
budget_ctrl	0.120	0.238	0.163	0.116	0.129	0.028	-0.470
hier_pomdp	0.120	0.191	0.129	0.097	0.140	0.033	-0.542
fail_aware	0.120	0.030	0.018	0.014	0.153	0.038	-0.769
robust_mpc	0.120	0.040	0.024	0.020	0.172	0.043	-0.787
option_term	0.120	0.000	0.000	0.000	0.188	0.060	-0.879
unc_halt	0.120	0.000	0.000	0.000	0.221	0.078	-0.970
retry	0.120	0.000	0.000	0.000	0.237	0.076	-0.996
safety_filter	0.120	0.000	0.000	0.000	0.255	0.097	-1.048
oracle	0.180	1.000	0.915	0.704	0.019	0.002	0.924
contain_v5	0.180	1.000	0.839	0.653	0.040	0.003	0.791
v4_graph	0.180	0.957	0.713	0.545	0.090	0.007	0.517
event_graph	0.180	0.719	0.481	0.372	0.125	0.018	0.081
budget_ctrl	0.180	0.647	0.431	0.316	0.129	0.028	-0.026
hier_pomdp	0.180	0.594	0.395	0.291	0.140	0.033	-0.106
robust_mpc	0.180	0.355	0.224	0.167	0.172	0.043	-0.452
fail_aware	0.180	0.311	0.207	0.148	0.153	0.038	-0.457
option_term	0.180	0.053	0.032	0.022	0.188	0.060	-0.822
unc_halt	0.180	0.018	0.010	0.006	0.221	0.078	-0.950
retry	0.180	0.000	0.000	0.000	0.237	0.076	-0.993
safety_filter	0.180	0.000	0.000	0.000	0.255	0.097	-1.046
oracle	0.240	1.000	0.915	0.704	0.019	0.002	0.925
contain_v5	0.240	1.000	0.839	0.653	0.040	0.003	0.791
v4_graph	0.240	1.000	0.746	0.568	0.090	0.007	0.568
event_graph	0.240	0.955	0.646	0.478	0.125	0.018	0.333
budget_ctrl	0.240	0.924	0.614	0.446	0.129	0.028	0.262
hier_pomdp	0.240	0.891	0.591	0.419	0.140	0.033	0.203
fail_aware	0.240	0.688	0.442	0.315	0.153	0.038	-0.069
robust_mpc	0.240	0.713	0.446	0.325	0.172	0.043	-0.089
option_term	0.240	0.401	0.238	0.168	0.188	0.060	-0.480
unc_halt	0.240	0.225	0.127	0.082	0.221	0.078	-0.755
retry	0.240	0.031	0.017	0.011	0.237	0.076	-0.964
safety_filter	0.240	0.016	0.008	0.005	0.255	0.097	-1.032
oracle	0.300	1.000	0.915	0.704	0.019	0.002	0.925
contain_v5	0.300	1.000	0.839	0.653	0.040	0.003	0.791
v4_graph	0.300	1.000	0.746	0.568	0.090	0.007	0.568
event_graph	0.300	1.000	0.675	0.497	0.125	0.018	0.377
budget_ctrl	0.300	1.000	0.665	0.480	0.129	0.028	0.338
hier_pomdp	0.300	0.995	0.659	0.461	0.140	0.033	0.307
fail_aware	0.300	0.939	0.602	0.421	0.153	0.038	0.182
robust_mpc	0.300	0.945	0.584	0.418	0.172	0.043	0.130
option_term	0.300	0.733	0.424	0.291	0.188	0.060	-0.174

Method	Budget	Coverage	Succ.	Contain	Cascade	Damage	Util.
unc_halt	0.300	0.571	0.318	0.198	0.221	0.078	-0.441
retry	0.300	0.310	0.174	0.098	0.237	0.076	-0.708
safety_filter	0.300	0.209	0.107	0.072	0.255	0.097	-0.864

G CITATION LEDGER

Table 14: Literature ledger used to keep the related-work boundary broad.
Boxed citation links route to bibliography entries.

#	Theme	References
1	temporal abstraction, options, and hierarchical reinforcement learning	(Sutton et al., 1999; Bacon et al., 2017; Nachum et al., 2018)
2	robot manipulation recovery and long-horizon execution	(Thananjeyan et al., 2021; Vats et al., 2024; Lin et al., 2025)
3	failure detection, anomaly attribution, and monitoring	(Zhang, 2026; Shah, 2025; Uranishi, 2026)
4	world models, task-and-motion planning, and sub-goal prediction	(Koike-Akino, 2026; Yoshie, 2024; Feng, 2026)
5	safety, uncertainty, risk, and calibration	(Choi, 2025; Wen, 2025; James, 2024)
6	embodied policy benchmarks and reproducibility	(Knightbridge, 2002; Russell, 2010; Azizan, 2025)
7	VLA failure recovery and executable repair policies	(Beksi, 2026; Grosu, 2025; Violante, 2022)
8	temporal abstraction, options, and hierarchical reinforcement learning	(Khorrami, 2025; Indurkha, 2026; Shkurti, 2025)
9	robot manipulation recovery and long-horizon execution	(Kantaros, 2023; Kim, 2026; Hashimoto, 2025)
10	failure detection, anomaly attribution, and monitoring	(Fiorini, 2020; Liu, 2025a; Fan, 2023)
11	world models, task-and-motion planning, and sub-goal prediction	(Lee, 2026; Huang, 2026; Jaber, 2026)
12	safety, uncertainty, risk, and calibration	(Singh; Liu, 2025b; Su, 2022)
13	embodied policy benchmarks and reproducibility	(Guo, 2024; Amato, 2025; Nguyen, 2024)
14	VLA failure recovery and executable repair policies	(Matsubara, 2024; Cong, 2026; Zhang, 2021)
15	temporal abstraction, options, and hierarchical reinforcement learning	(Piater, 2016; Kantaros, 2024; Hein, 2023)
16	robot manipulation recovery and long-horizon execution	(Kim, 2025; Fern, 2026; Li, 2026b)
17	failure detection, anomaly attribution, and monitoring	(Weiger, 2022; Tomizuka, 2021; Umar, 2025)
18	world models, task-and-motion planning, and sub-goal prediction	(Haddadin, 2017; Ribas-Xirgo, 2026; Gandolla, 2023)
19	safety, uncertainty, risk, and calibration	(Cao, 2021; Johal, 2025; Liu, 2024)

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#	Theme	References
20	embodied policy benchmarks and reproducibility	(Brandao, 2025; Scassellati, 2016; Tasaki, 2025)
21	VLA failure recovery and executable repair policies	(Rodriguez-Ayerbe, 2022; Seita, 2025; Schoellig, 2026)
22	temporal abstraction, options, and hierarchical reinforcement learning	(del Cerro; Antonio Barrientos, 2020; Kroemer, 2021; Short, 2024)
23	robot manipulation recovery and long-horizon execution	(Xu, 2026; Haddadin, 2016; Harada, 2016)
24	failure detection, anomaly attribution, and monitoring	(Zhao, 2022; Rao, 2012; Ding, 2023)
25	world models, task-and-motion planning, and sub-goal prediction	(Cui, 2022; Howard, 2021; Pesty, 2021)
26	safety, uncertainty, risk, and calibration	(Bruyninckx, 2023; Su, 2024; Harada, 2019)
27	embodied policy benchmarks and reproducibility	(Ahn, 2023; Weldegioris, a; Sunderhauf, 2022)
28	VLA failure recovery and executable repair policies	(Chai, 2021; Kulic, 2021; Pham, 2023)
29	temporal abstraction, options, and hierarchical reinforcement learning	(D’Andrea, 2022; Marchionni, 2025; Weldegioris, b)
30	robot manipulation recovery and long-horizon execution	(Singh, 2005; Chen, 2021; Kragic, 2008)
31	failure detection, anomaly attribution, and monitoring	(Lee, 2023; Jeong, 2024; Taniguchi, 2024)
32	world models, task-and-motion planning, and sub-goal prediction	(Fu, 2023; Liu, 2023; Chen, 2019)
33	safety, uncertainty, risk, and calibration	(Shi, 2020; Jenkins, 2022; Li, 2026a)
34	embodied policy benchmarks and reproducibility	(Li, 2025; Sun, 2024; Schaal, 2022)
35	VLA failure recovery and executable repair policies	(Jagersand, 2024; Oltramari, 2024; Guo, 2023)

H REPRODUCIBILITY CHECKLIST

- The frozen plan is stored in `docs/paper105-expanded-submission-plan-20260622.md`.
- The final runner is `src/run_experiment.py`; it streams raw rollouts and writes machine-readable summaries.
- The manuscript is generated by `scripts/generate_manuscript.py`; empirical claims are drawn from CSV/JSON outputs.
- The validation script checks row counts, finite CSV values, boxed citation-link settings, canonical PDF placement, page count, and terminal scope status.
- No PDF is copied to the visible Desktop.
- The canonical PDF is `C:/Users/wangz/Downloads/105.pdf`.

I REVIEWER ATTACK MATRIX

Table 15: Hostile-review attacks and the corresponding evidence hook.

Attack	Evidence hook
Hierarchy already solves this.	Hierarchy-only and option-termination baselines are included; both lose under cascade and budget stress.
This is just a safety filter.	The local safety filter is strong locally but accumulates subgoal and state corruption under cross-level failures.
Retry/recovery is enough.	Reactive retry and recovery are tested and fail when local repair silently corrupts task state.
The method wins by over-halting.	False-halt rate, coverage, and fixed-risk utility are reported; no false_halt_calibration is a near ablation.
The v4 graph already had the mechanism.	v4 is the strongest non-oracle baseline and is still beaten by v5 on success, containment, and utility.
The result is a lucky seed.	Pairwise seed tests are reported against every baseline.
The oracle gap is hidden.	The oracle is included in main, stress, and fixed-risk results and remains above v5.
The benchmark is synthetic.	Correct; the scope gate stays false and the terminal decision is STRONG_REVISE.
The tables are cherry-picked.	Full split, full stress, full fixed-risk, row-count, and negative-case ledgers are appended.
Ablations are decorative.	Every named mechanism is removed under the same frozen hard-split protocol.
Fixed-risk results are abstention gaming.	Coverage is reported and abstention counts as zero success.
The literature boundary is too narrow.	The generator filters a local robotics/failure/hierarchy pool and exposes a citation ledger.
The page count is padding.	Appendices are row counts, protocol loads, full results, attack matrix, and reproducibility notes.
It is not ICLR-main ready.	Agreed: no real robot or independent high-fidelity validation exists yet.

J GATE INTERPRETATION NOTES

Success gate. The success gate is intentionally defined against the strongest non-oracle reference rather than a weak behavior-cloning baseline.

Containment gate. Containment is reported separately from success because a method can complete some tasks while still allowing subgoal or task-state corruption.

Cascade and corruption gates. Cascade, state corruption, and subgoal corruption are separate because a low-level anomaly may not damage the physical scene but can still invalidate future high-level decisions.

False-halt and missed-failure gates. These two metrics expose opposite failure modes. A conservative method can over-halt valid skills; an aggressive method can miss real failures. The paper reports both.

Fixed-risk gate. The fixed-risk protocol consumes a predicted containment risk. Coverage is shown so abstention cannot masquerade as safe success.

Scope gate. The scope gate is the honesty firewall. Local synthetic evidence can justify a strong-revise artifact, but it cannot establish real robot readiness.

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